Mountain Waves in the Sky: Overshooting Top Dynamics above Supercell Thunderstorms

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August 2026

MSc Thesis

Final Report

Abstract

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Acknowledgements

I would like to express my gratitude to my supervisor Morgan O'Neill for ....

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1 Introduction

Stratospheric water vapour (SWV) is an important climate feedback, with the ability to potentially impact surface warming [1], hamper ozone recovery [2] and decrease the intensity of the stratospheric circulation [3], influencing the global ozone distribution. Thus, it is critical to understand how the SWV budget in order to quantify these feedbacks. Unfortunately, global climate models currently struggle with this, with an average 0.5 ppmv underestimation of ppmv at 70hPa alongside a large model spread [4]. An important secondary source of SWV is mid-latitude convection from supercell thunderstorms. These storms produce overshooting tops (OTs), deep convection which overshoots the level of neutral buoyancy and penetrates up into the stratosphere. With strong storm-relative prevailing winds in the stratosphere large amounts of water vapour within this OT can be injected irreversibly into the stratosphere in the form of a turbulent hydraulic jump [5]. However, this hydraulic jump is a relatively new discovery and, as of yet, very few studies have taken to analyzing it.

1.1 AACP Overview

The SWV feedback on surface temperature has been a source of contention in recent times. In 2001, [2] found that the combined water vapour - ozone feedback due to changes from 1959 to 1999 led to a net 0.07 Kelvin surface warming per decade. This was despite the ozone feedback on its own having a net cooling effect. In 2013, [1] demonstrated that increases in SWV due to increases in tropospheric temperature causes a $0.3W/m^2$ increase in surface flux per Kelvin of surface warming. A third of this was found to be due to water vapour entering the stratosphere through the tropical tropopause layer, with the remaining two thirds being due to water vapour entering through extra-tropics. However, recent studies challenge this claim. For example, using a multi-model analysis, He at al. (2025) [6] claimed that after both stratospheric cooling and tropospheric warming feedbacks associated with SWV changes are included, the total radiative forcing due to SWV is negligible. The uncertainty around the radiative feedback of SWV further increases the importance of accurately calculating its budget, and finding how it will change under climate change, for future studies.

Water vapour can travel up into the stratosphere in many different ways. The main method of transport is through the tropical cold point tropopause [4], although transport from volcanic eruptions [7] and convective overshooting are important secondary sources.

Focusing on convective overshooting, Dauhut et al. (2018) [8] used a model to show that tropical convective storms, such as the "Hector the Convector" can inject over 2700 tons of water vapour into the stratosphere over the course of an hour, with this injection being poorly resolved by convective parameterizations. In the midlatitudes, an aircraft flown 20 hours after a supercell thunderstorm showed that large 1-2 ppmv enhancements in water vapour were found well above the tropopause (19.25 km up!) [9]. Additionally, model simulations have shown that water vapour injected into the stratospheric overworld (above the 380K potential temperature isoline) by OTs tends to stay even after the supercell system which triggered them is gone [10]. This injection is not well resolved by current generation satellites such as Aura's Microwave Limb Sounder.

In about 10% of cases, storms produce an above anvil cirrus plume (AACP), with AACP's being formed by 75% of supercells. Despite this, AACPs produce 57% of all severe weather reports and 73% of significant severe weather reports [11]. AACP's are characterized by a unique texture distinct from the storm larger anvil several kilometers underneath, as well as as a high IR brightness temperature (but not always)! AACPs precede severe weather reports by 31 minutes on average, and this one feature predates U.S National Weather Service warnings 25% of the time. It has also been shown that AACP storms inject more water into the stratospheric overworld [5] [12] [13], although whether they inject more into the stratospheric middleworld is disputed. High storm-relative winds in the lower stratosphere and increasing low-level instability have been found to be environmental conditions which favour AACP formation [13].

The mechanism behind AACP formation is the breaking of gravity waves forced by the OT [5] [8] [14] [15] [16]. The OT acts similarly to a mountain at the surface, blocking the incoming stratiform flow, and forcing the winds up and over the OT. The flow then accelerates over this "mountain", continuing to accelerate as it travels over the leeside of the OT. These waves break, and the accelerated flow reconnects to the ambient downstream flow in the form of a turbulent hydraulic jump [5]. This hydraulic jump takes the high levels of water perturbation from the OT (which originated in the upper troposphere [15]) and mixes it into the stratosphere (see O'Neill et al. (2021) [5] figure 2B). The cloud ice "jumped" from the OT is what forms the AACP. In subsaturated environments, this cloud ice will then sublimate, irreversibly adding water vapour into the stratosphere. However, in supersaturated environments the vapour will actually deposit on the ice crystals causing the stratosphere to lose moisture when the ice precipitates out [15]. When the storm ends, a lot of this water vapour which hasn't been mixed in this fashion will fall back into the troposphere with the OT [10]. In their 1km grid spacing simulation, Qu et al. (2020) [16] found that wave breaking accounted for 59% of the vertical vapour injection via vertical transport while the mean updraft accounted for 39%. However, this ratio is expected to skew further towards wave breaking as the simulation resolution is increased and more wave breaking is resolved.

1.2 Mountain Waves

With the overshooting top acting as a blockage to oncoming stratiform flow, it stands to reason that theory describing fluid flow over other topography can be used. This has been studied since at least the 1950s [17], with a significant bulk of research completed on the topic since then.

A lot of this research has involved shallow water flow, and while this is a simpler case then the stratified flow seen in the atmosphere, it can still be used to better understand the theory. With this, the steady state shallow water equations can be used to describe the flow [18]:

$$u\frac{\partial u}{\partial x} + g\frac{\partial D}{\partial x} + g\frac{\partial h}{\partial x} = 0, \tag{1}$$

$$\frac{\partial uD}{\partial x} = 0. \tag{2}$$

Here, D is the fluid depth and h is the obstacle height. Equation 1 describes momentum continuity, where momentum via horizontal advection (the first term) of the fluid is balanced by gravitational potential from either the fluid piling up with increased fluid depth (second term) or as the fluid rising as it is forced over the obstacle (third term). Equation 2 describes mass continuity where changing fluid velocity over x must be accompanied by a corresponding change in the fluid depth. Manipulating we see that [19]:

$$(1 - \mathrm{Fr}^2)\frac{\partial D}{\partial x} = -\frac{\partial h}{\partial x}, \tag{3}$$

where Fr is the Froude number, and in this case is U/gD. From equation 3, we can see that if the upstream Froude number is less than one (middle of Fig 1), mountain height increases the fluid depth must decrease. As fluid depth decreases, the fluid speed increases over the mountain by mass continuity. Similarly, for an upstream Froude number over one (top of Fig 1), the fluid acts like a ball going over a hill, slowing down and thickening as it goes over the mountain before accelerating down the other side. However, in a certain regime of subcritical upstream Froude number, with a tall enough mountain (see Fig 2.9 of Baines (2022) [20]), the fluid accelerates enough to let the Froude number transition from subcritical to supercritical over the mountain (bottom of Fig 1). This causes the fluid

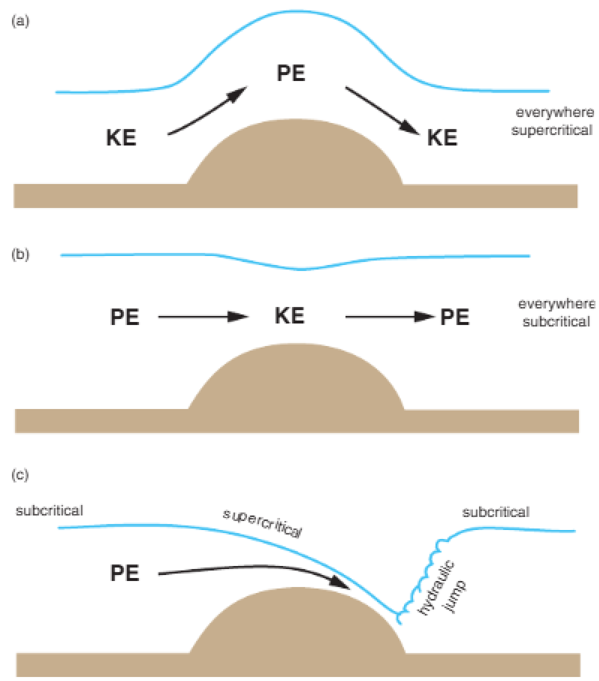

Figure 1: Figure 12.12 from Markowski and Richardson (2011) [19]. Flow over an obstacle for a single layer fluid. Top panel is for supercritical flow everywhere (Fr ¿ 1), the middle panel is for subcritical flow everywhere (Fr ¡ 1), and the bottom panel shows a transition from subcritical to supercritical over the barrier.

accelerating all the way over and down the mountain, creating a downslope windstorm. The fluid then recovers to the ambient flow in the form of a turbulent hydraulic jump. However, this Froude number definition is suspect at best for an atmospheric context [19] as the depth of the layer is not well defined for stratified flows.

In a simulation of two layer flow with a transition layer past topography, Rotunno and Bryan (2018) [21] [22] show that steepening of leeside isentropes baroclinically produces vorticity. This baroclinically produced vorticity is transported further downstream where it causes the isentropes to overturn. This isentropic overturning causes a location of local instability, the wave breaks, and the transition layer mixes in with itself forming a hydraulic jump. The results of this simple surface simulation (figure 3e of [21]) shows a remarkable resemblance to the result given in O'Neill et al (2021) ([5] fig 2B). This demonstrates that this surface mountain wave theory is indeed quite applicable to mountain waves in the sky.

A hydraulic jump can be thought of as a discontinuity between supercritical and subcritical flows. This is demonstrated using the QRS framework in Baines (2022) [20]. Here Q represents mass flux, R represents streamline energy (or the Bernoulli function) and S represents the momentum flux. A phase diagram of normalized S and normalized R (fig 2.20 of Baines (2022) [20]) can be created, with the upper edge of the drawn cusp representing subcritical flow and the lower edge being supercritical flow. The area within those edges represent periodic wave solutions which tend to zero amplitude as we move closer to the subcritical branch. A hydraulic jump is a horizontal line along the cusp of that figure, with the momentum being constant while energy is lost due to turbulence and viscosity. However, if R is decreased by an energy less than that required to travel across the gap, we get a wave-train. For supercritical Froude numbers less than 1.3, a tiny amount of dissipation converts the flow into large amplitude undular bores. For Froude numbers over 1.3 however, we need a lot of energy dissipation to cross an unsteady zone on the phase diagram, leaving a more turbulent structure as the leading wave breaks. For Froude numbers over about 1.55, we get a fully turbulent bore. The bore amplitude increases as we increase the Froude number from one, until the 1.3 value is hit and the leading wave breaks. Afterwards, the amplitude decreases with increasing Froude number. However, the jump amplitude for any hydraulic jump monotonically increases with Froude number.

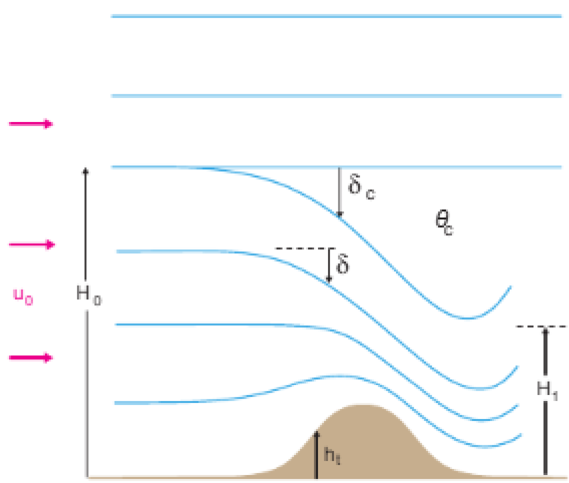

Figure 2: Figure 12.13 from Markowski and Richardson (2011) [19]. A depiction of the dividing streamline where δ_c represents the height deviation of the lower branch of the streamline from it's upstream height (H_0).

For the stratified infinite depth flows seen in the atmosphere, Smith (1985) [23] devised a theory where when standing waves overturn/break, there is a region with low stability and a critical level. A critical level occurs when the phase speed of a wave equals the wind speed, causing group velocity to become nearly horizontal and is associated with wave reflection [19]. Smith (1985) [23] specified a dividing streamline 2, which, for a tall enough mountain, descends far down towards the surface. The region below the streamline is described by steady flow, but within it is a well mixed region of constant θ. The critical layer does not allow wave energy to propagate through the mixed layer, reflecting the wave energy back down causing high speed downslope windstorms. If the dividing streamline is within a certain number of wavelengths off the ground, there is a hydraulic jump.

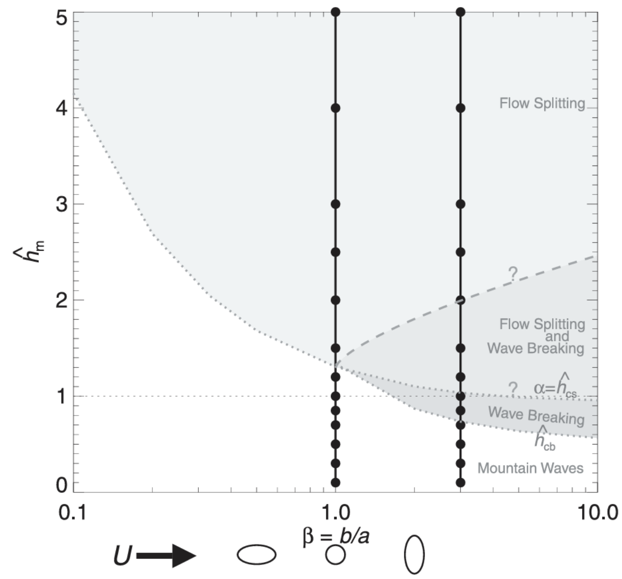

Figure 3: Figure 1 from Eckermann et al. (2010) [24]. A phase space diagram of different flow regimes for different mountain heights and shapes.

Additionally, the shape of the mountain also affects the flow type as seen in figure 3 [24]. In figure 3, $\hat{h}_m$ is the normalized mountain height equal to Nh_m/U, where U is the upstream fluid velocity, N is the Brunt–Väisälä frequency and h_m is the peak mountain height. Note that $\hat{h}_m$ is often interpreted as an inverse Froude number for these stratified flows. However, as expressed in Baines (2022) [20], there are many different Froude numbers choose from in this context. Returning to figure 3, wave breaking only occurs for values of β larger than one and for taller normalized mountain heights. However, if the mountain is too tall, the waves won't break and the flow will simply split over either side of the mountain.

1.3 Current Work

Unfortunately, despite the large amounts of observational and modeling studies on OTs and the vast array of literature pertaining to mountain waves, very few studies have done in depth dynamics modeling of OTs using mountain wave results. One of the few such studies, O'Neill et al. (2021) [5], had an extremely idealized environment for their simulation. They used updraft nudging for their initial convective initiation, alongside an idealized supercell thunderstorm sounding and a horizontally homogeneous domain. This is despite their extremely high resolution simulation, with grid spacing in all three dimensions of 50m. Nevertheless, the idealities of their simulation likely led to some artifacts, such as cloud ice suspended above the OT before the gravity wave break. This cloud ice could potentially provide a critical layer for gravity wave reflection; hence, it is imperative to understand whether it is an artifact, and if it's not, to understand what its source was.

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